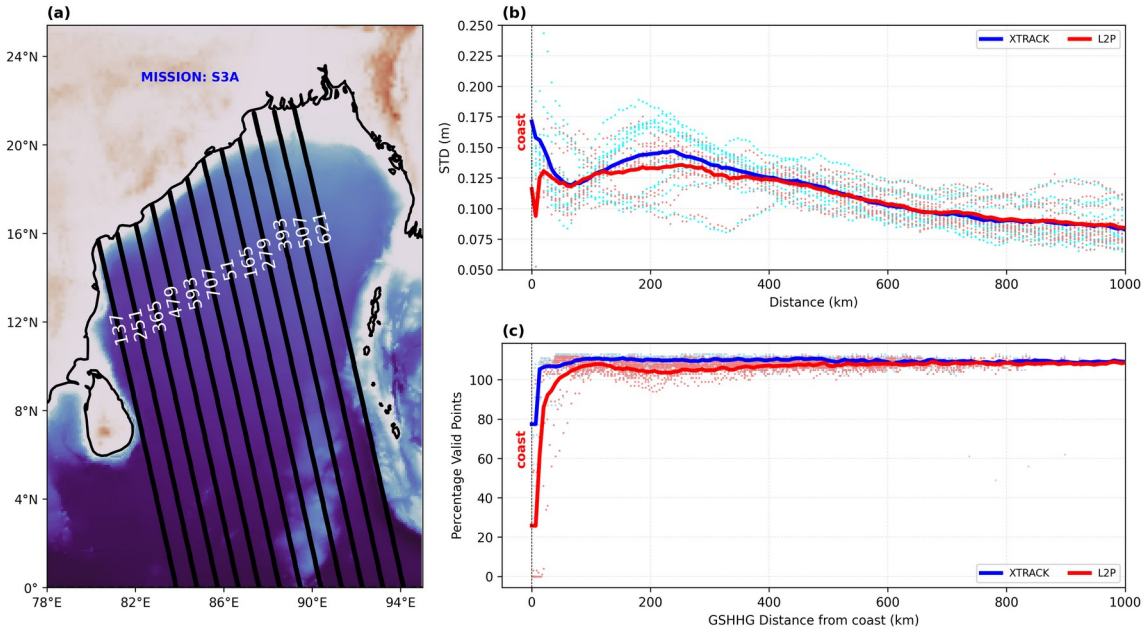


Fig1: Comparison of Sentinel-3A (S3A) sea level anomaly (SLA) data between X-TRACK and along-track Level-2+ (L2P) products for common ground tracks in the North Indian Ocean. (a) Spatial distribution of selected common S3A passes over ETOPO5 bathymetry, with track numbers annotated in red. (b) Temporal standard deviation of SLA as a function of distance from coast for individual tracks (dots) and their multi-track mean (solid lines) from X-TRACK (blue) and L2P (gray). (c) Number of valid SLA observations per along-track point plotted against GSHHG coastline distance, showing X-TRACK (blue) and L2P (gray) individual track values (dots) and multi-track means (solid lines). The dashed vertical line in (b) and (c) marks the coastline

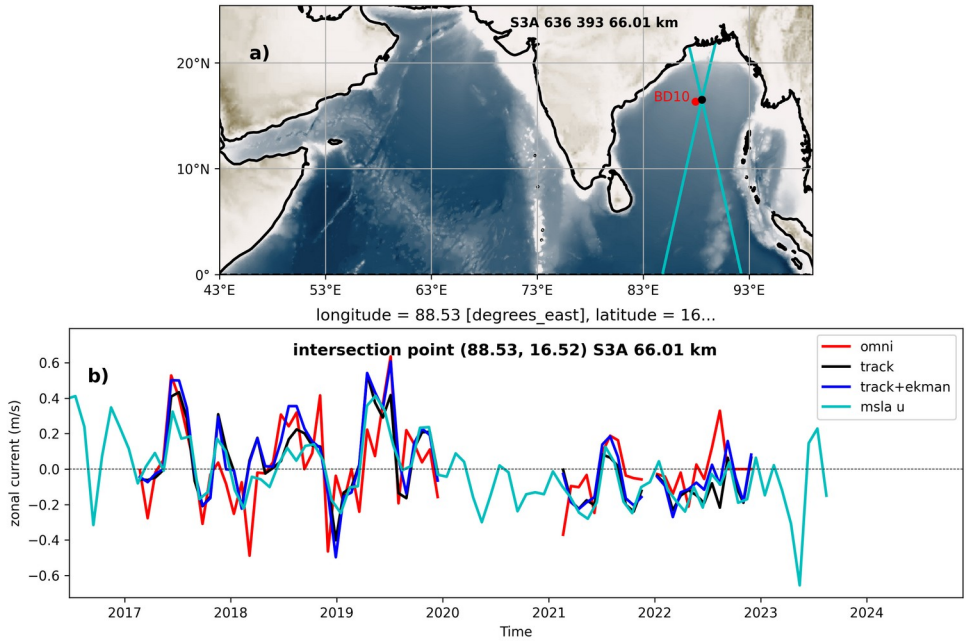


Fig2: b) Time series of geostrophic zonal component (black) derived at an intersection point which is close to an OMNI buoy (BD10) overlied with buoy's zonal current(red). Timeseries of gu correted with ekman u component is also added (blue). a) shows the intersection point(black) of the tracks 636 and 393 and close by bouy(red). It is rougly half a degree away from the intersection point.

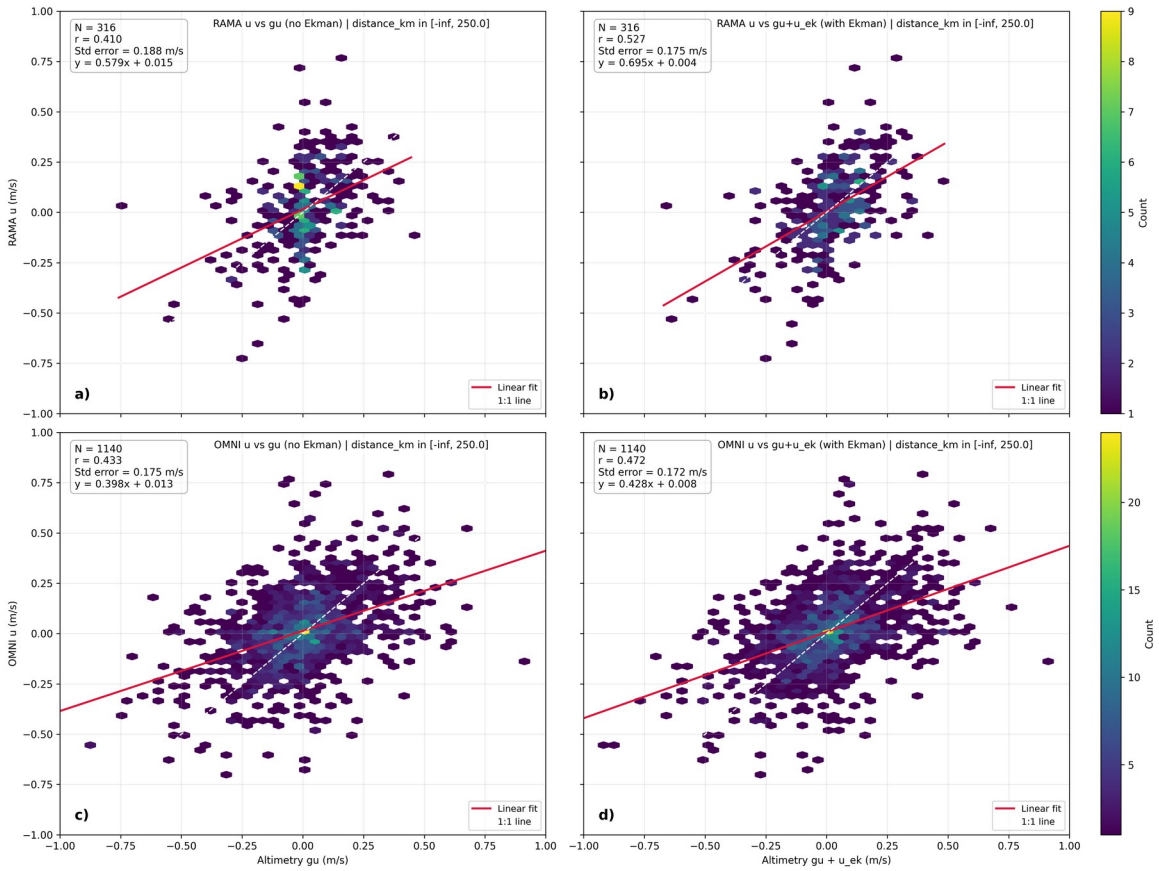


Fig3: a) Scatter plot of rama buoys' zonal currents(y-axis) versus altimetry derived zonal currents(x-axis) at nearby intersection points. b) Shows scatter plot with altimetry derived currents corrected with ekman zonal currents. c) and d) show scatter plots of OMNI currents with and without Ekman correction against altimetry zonal currents. The colorbar shows density of points in the scatter plot in terms of count. A redline passing through the points indicates linear fit.

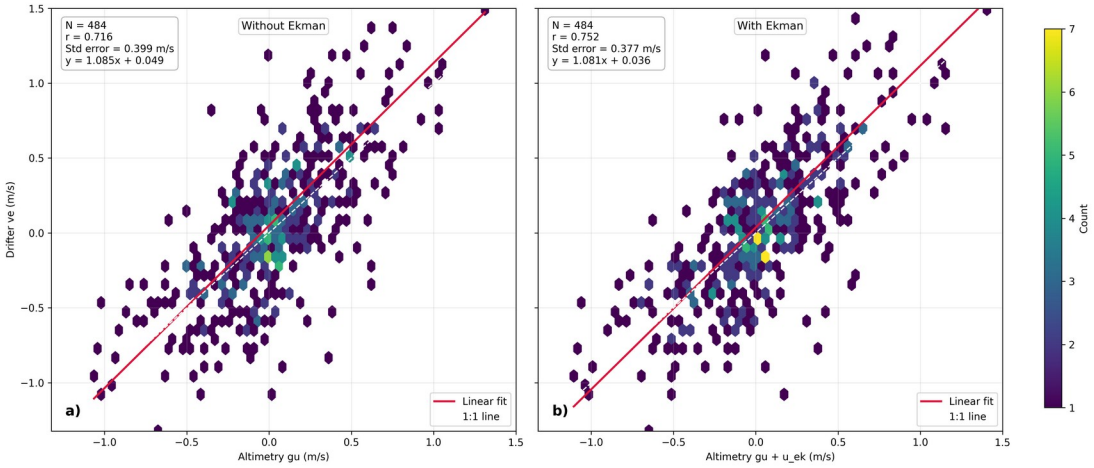


Fig4: a) Scatter plot of drifters' zonal velocities versus altimetry derived velocities within 200km inshore of the coast for the satellite TP+J1+J2+J3+S6A. b) shows scatter plot with altimetry derived velocities corrected with ekman zonal velocities.

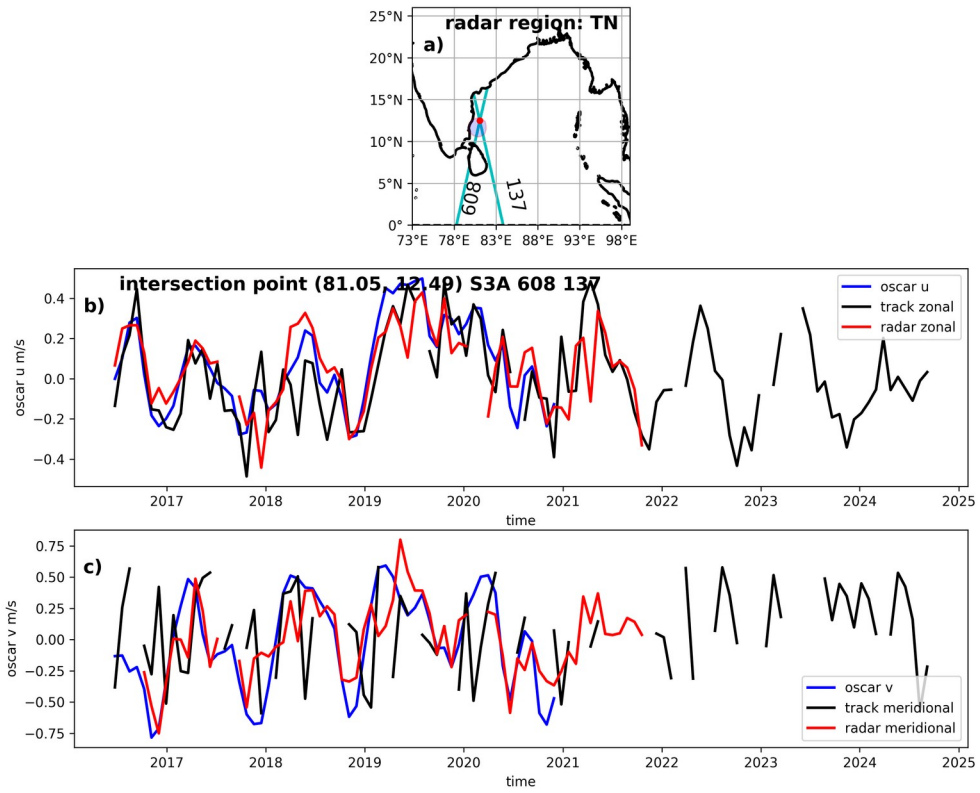


Fig5: a) Shows the radar region of TN with a intersection point within the region. b) and c) show zonal and meridional currents of altimetry derived geostrophic currents at the intersection point along with radar zonal velocity time series.

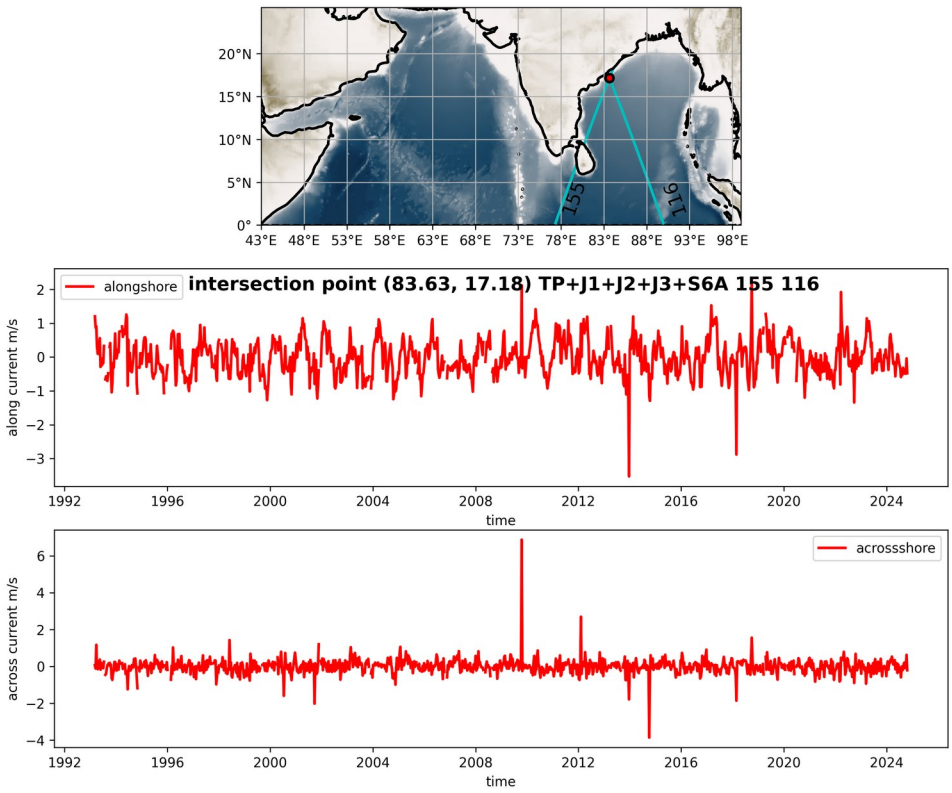


Fig6: a) An intersection point very close to the east coast of India belonging to the satellite mission TP+J1+J2+J3+S6A. b) and c) show thirty year timeseries of along shore and across shore current components at the intersection of two satellite tracks (155, 116).

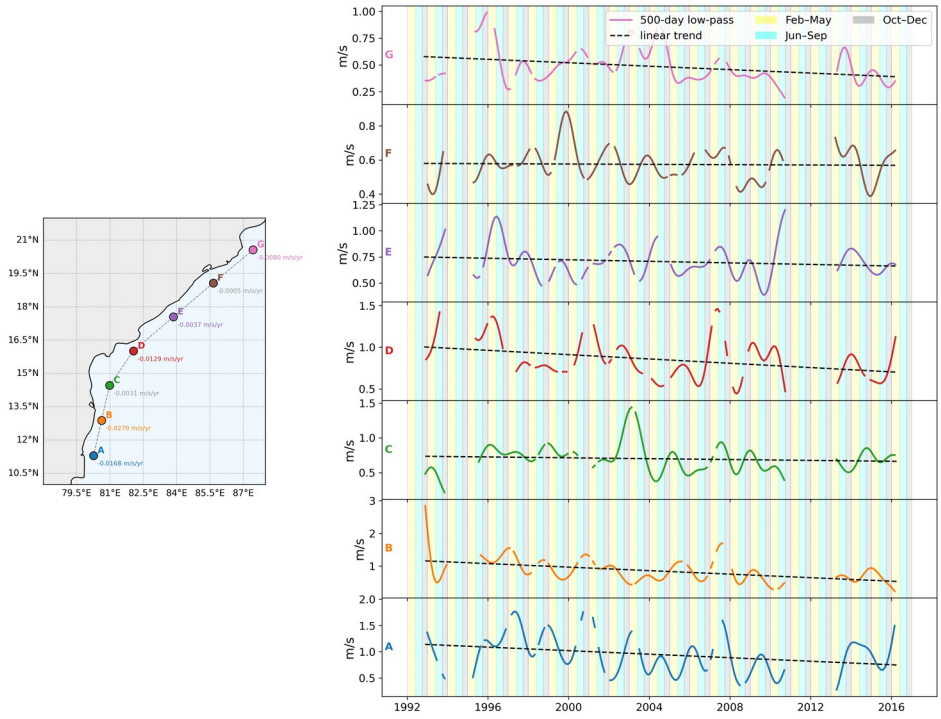


Fig7: Fig. X. Long-term variability and linear trends of geostrophic current speed at seven crossover points of the East India Coastal Current (EICC) derived from ERS-1/2, Envisat, and SARAL/AltiKa altimetry. (Left) Locations of the seven crossover points (A–G, south to north) along the eastern boundary of the Bay of Bengal (10°–22°N, 78°–88°E). Each point is colour-coded by its label; the annotated value is the linear trend in m s^{-1} yr^{-1} (greyed out where the trend is not significant at the 95% confidence level, $p \geq 0.05$). (Right) 500-day low-pass filtered current speed (coloured line) and its linear trend fit (black dashed line) at each point. Background shading indicates the spring inter-monsoon (February–May, yellow), summer monsoon (June–September, cyan), and post-monsoon/early winter (October–December, grey) seasons. Point coordinates and contributing ground-track pairs are: A (80.26°E, 11.30°N; T868×T567), B (80.62°E, 12.89°N; T868×T23), C (80.98°E, 14.46°N; T868×T481), D (82.06°E, 16.01°N; T324×T395), E (83.86°E, 17.55°N; T238×T767), F (85.65°E, 19.07°N; T152×T137), and G (87.45°E, 20.57°N; T66×T509).